Continuous-Time Almgren–Chriss Model

Implementation Shortfall, Target Close, and TWAP

Static Mean–Variance Strategies, No Permanent Impact

1 Model Setup

We want to liquidate a block of $Q > 0$ shares over a fixed time horizon $[0, T]$. The key quantities are:

- q_t: the number of shares still held (*inventory*) at time t.
- $v_t = -\dot{q}_t$: the *selling speed*, i.e. how many shares we sell per unit of time. A positive v_t means we are selling.
- Boundary conditions: $q_0 = Q$ (we start with everything) and $q_T = 0$ (we must finish flat). These two conditions together imply

$$\int_0^T v_t \, \mathrm{d}t = Q.$$

Price dynamics (no permanent impact). The unaffected mid-price follows a Brownian motion with no drift:

$$\mathrm{d}S_t = \sigma \, \mathrm{d}W_t, \qquad S_t = S_0 + \sigma W_t,$$

where $\sigma > 0$ is the volatility and W_t is a standard Brownian motion.

Temporary market impact. When we sell at speed v_t, we push the price against ourselves momentarily. The actual execution price is

$$\tilde{S}_t = S_t - \eta v_t,$$

where $\eta > 0$ is the temporary impact parameter. Selling $v_t \, \mathrm{d}t$ shares at this price generates cash:

$$\mathrm{d}X_t = \tilde{S}_t \, v_t \, \mathrm{d}t = (S_t - \eta v_t) \, v_t \, \mathrm{d}t.$$

Total cash received. Integrating over the whole horizon:

$$X_T = \int_0^T (S_t - \eta v_t) \, v_t \, \mathrm{d}t.$$

Criterion. We use a *mean–variance* objective: for risk-aversion parameter $\lambda > 0$, we minimise

$$J = -\mathbb{E}[\Pi] + \lambda \operatorname{Var}(\Pi),$$

where Π is the *payoff* of the strategy relative to a chosen benchmark. We restrict to *static* (deterministic) strategies: q_t and v_t are deterministic functions of time, so they are not adapted to the Brownian motion.

2 A Shared Tool: Integration by Parts on the Cash Flow

Before treating each benchmark separately, we derive a representation of X_T that will be used throughout.

Expanding the cash flow. Substituting $S_t = S_0 + \sigma W_t$:

$$\begin{aligned} X_T &= \int_0^T S_t\, v_t \,\mathrm{d}t - \eta \int_0^T v_t^2 \,\mathrm{d}t \\ &= S_0 \int_0^T v_t \,\mathrm{d}t + \sigma \int_0^T W_t\, v_t \,\mathrm{d}t - \eta \int_0^T v_t^2 \,\mathrm{d}t. \end{aligned} \tag{1}$$

Using the boundary condition $\int_0^T v_t \,\mathrm{d}t = Q$, the first term simplifies to $S_0 Q$:

$$X_T = S_0 Q + \sigma \int_0^T W_t\, v_t \,\mathrm{d}t - \eta \int_0^T v_t^2 \,\mathrm{d}t. \tag{2}$$

Stochastic integration by parts. The stochastic integral $\int_0^T W_t\, v_t \,\mathrm{d}t$ is an ordinary (Lebesgue) integral because v_t is deterministic, but W_t is random. We convert it into a stochastic integral (with respect to W) using integration by parts. Recall that $v_t = -\dot{q}_t$, so $v_t \,\mathrm{d}t = -\,\mathrm{d}q_t$. We integrate by parts with

$$u = W_t, \quad \mathrm{d}V = v_t \,\mathrm{d}t = -\,\mathrm{d}q_t :$$

$$\int_0^T W_t\, v_t \,\mathrm{d}t = -\int_0^T W_t \,\mathrm{d}q_t = -\big[W_t\, q_t\big]_0^T + \int_0^T q_t \,\mathrm{d}W_t.$$

Evaluating the boundary term: $W_0 = 0$ and $q_T = 0$, so

$$\big[W_t\, q_t\big]_0^T = W_T \cdot 0 - 0 \cdot Q = 0.$$

Therefore:

$$\boxed{\int_0^T W_t\, v_t \,\mathrm{d}t = \int_0^T q_t \,\mathrm{d}W_t.} \tag{3}$$

The left-hand side of (3) mixes the deterministic function v_t with the Brownian motion W_t in a way that is hard to work with directly. The right-hand side is a *stochastic integral* driven by $\mathrm{d}W_t$ with the **deterministic** integrand q_t. For a

deterministic integrand, Itô's isometry gives directly:

$$\mathrm{Var}\left(\int_0^T q_t \,\mathrm{d}W_t\right) = \int_0^T q_t^2 \,\mathrm{d}t.$$

This is the key that unlocks the variational formulation.

Canonical form of the cash flow. Substituting (3) into (2):

$$X_T = S_0 Q - \eta \int_0^T v_t^2 \,\mathrm{d}t + \sigma \int_0^T q_t \,\mathrm{d}W_t. \tag{4}$$

This form holds *for any admissible strategy* and will be our starting point in each of the three benchmark cases below.

3 Implementation Shortfall (IS)

3.1 Benchmark and payoff

The *implementation shortfall* benchmark is the initial price S_0: the ideal would be to sell all Q shares instantly at S_0, receiving QS_0. The shortfall is the gap between this ideal and what we actually collect:

$$\Pi_{\mathrm{IS}} = X_T - QS_0.$$

Minimising the shortfall (in mean–variance sense) is equivalent to maximising the cash X_T, so we use the objective

$$J_{\mathrm{IS}} = -\mathbb{E}[\Pi_{\mathrm{IS}}] + \lambda \,\mathrm{Var}(\Pi_{\mathrm{IS}}).$$

Since QS_0 is a constant, it vanishes from both the mean and the variance: $\mathbb{E}[\Pi_{\mathrm{IS}}] = \mathbb{E}[X_T] - QS_0$ and $\mathrm{Var}(\Pi_{\mathrm{IS}}) = \mathrm{Var}(X_T)$.

3.2 Mean of X_T

From the canonical form (4), the only random term is $\sigma \int_0^T q_t \,\mathrm{d}W_t$. Since q_t is deterministic and $\mathbb{E}[\,\mathrm{d}W_t] = 0$, the stochastic integral has zero expectation. Therefore:

$$\mathbb{E}[X_T] = S_0 Q - \eta \int_0^T v_t^2 \,\mathrm{d}t.$$

The term $-S_0 Q$ is a constant and can be dropped from the optimisation, leaving:

$$-\mathbb{E}[\Pi_{\mathrm{IS}}] = \eta \int_0^T v_t^2 \,\mathrm{d}t = \eta \int_0^T \dot{q}_t^2 \,\mathrm{d}t.$$

3.3 Variance of X_T

The deterministic terms in (4) do not contribute to variance. Applying Itô's isometry to the stochastic integral:

$$\mathrm{Var}(X_T) = \sigma^2 \,\mathrm{Var}\left(\int_0^T q_t \,\mathrm{d}W_t\right) = \sigma^2 \int_0^T q_t^2 \,\mathrm{d}t.$$

3.4 Variational formulation

Combining mean and variance:

$$J_{\mathrm{IS}}[q] = \eta \int_0^T \dot{q}_t^2 \, \mathrm{d}t + \lambda\sigma^2 \int_0^T q_t^2 \, \mathrm{d}t, \tag{5}$$

subject to $q_0 = Q$, $q_T = 0$.

The first term $\eta \int \dot{q}_t^2$ is the *impact cost*: selling faster (larger $|\dot{q}_t|$) costs more. The second term $\lambda\sigma^2 \int q_t^2$ is the *market risk*: holding more inventory (larger $|q_t|$) exposes us to more price uncertainty. The optimal schedule balances these two forces.

3.5 Euler–Lagrange equation

The Lagrangian of (5) is $\mathcal{L}(q, \dot{q}) = \eta\dot{q}^2 + \lambda\sigma^2 q^2$. The Euler–Lagrange equation $\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right) - \frac{\partial \mathcal{L}}{\partial q} = 0$ gives:

$$\frac{d}{dt}(2\eta\dot{q}) - 2\lambda\sigma^2 q = 0 \implies \ddot{q} - \kappa^2 q = 0, \qquad \kappa^2 := \frac{\lambda\sigma^2}{\eta}.$$

3.6 Solution

The general solution of $\ddot{q} - \kappa^2 q = 0$ is:

$$q(t) = A\cosh(\kappa t) + B\sinh(\kappa t).$$

Applying the boundary conditions:

At $t = 0$: $q(0) = A = Q$, so $A = Q$.

At $t = T$: $Q\cosh(\kappa T) + B\sinh(\kappa T) = 0$, so $B = -Q\dfrac{\cosh(\kappa T)}{\sinh(\kappa T)} = -Q\coth(\kappa T)$.

Therefore $q(t) = Q\cosh(\kappa t) - Q\coth(\kappa T)\sinh(\kappa t)$. Using the hyperbolic identity $\sinh(\kappa(T-t)) = \sinh(\kappa T)\cosh(\kappa t) - \cosh(\kappa T)\sinh(\kappa t)$, we can rewrite this more compactly as:

$$q_{\mathrm{IS}}(t) = Q\,\frac{\sinh\big(\kappa(T-t)\big)}{\sinh(\kappa T)}, \qquad v_{\mathrm{IS}}(t) = Q\,\frac{\kappa\cosh\big(\kappa(T-t)\big)}{\sinh(\kappa T)}, \qquad \kappa^2 = \frac{\lambda\sigma^2}{\eta}.$$

For large κ (high risk aversion or high volatility), the schedule front-loads selling: q_t drops quickly then flattens. For small κ, it is close to linear.

4 Target Close (TC)

4.1 Benchmark and payoff

The *target close* benchmark is the terminal price S_T: we measure our performance relative to the price at the end of the trading window. The payoff is:

$$\Pi_{\mathrm{TC}} = X_T - QS_T.$$

4.2 Simplifying the payoff

Starting from the canonical form (4) and writing $S_T = S_0 + \sigma W_T$:

$$\Pi_{\mathrm{TC}} = S_0 Q - \eta \int_0^T v_t^2 \,\mathrm{d}t + \sigma \int_0^T q_t \,\mathrm{d}W_t - Q(S_0 + \sigma W_T).$$

The $S_0 Q$ terms cancel. We express $W_T = \int_0^T \mathrm{d}W_t$, so $QW_T = Q \int_0^T \mathrm{d}W_t$:

$$\begin{aligned}\Pi_{\mathrm{TC}} &= -\eta \int_0^T v_t^2 \,\mathrm{d}t + \sigma \int_0^T q_t \,\mathrm{d}W_t - Q\sigma \int_0^T \mathrm{d}W_t \\ &= -\eta \int_0^T v_t^2 \,\mathrm{d}t + \sigma \int_0^T (q_t - Q) \,\mathrm{d}W_t. \end{aligned} \tag{6}$$

Comparing (6) with the IS expression, the only difference is that the stochastic integrand changes from q_t to $q_t - Q$. This reflects the change of benchmark: for TC we care about how much q_t deviates from Q (i.e. how much we have *already sold*), whereas for IS we care about q_t itself (how much we still *hold*).

4.3 Mean and variance

Mean. The stochastic integral has zero expectation:

$$-\mathbb{E}[\Pi_{\mathrm{TC}}] = \eta \int_0^T v_t^2 \,\mathrm{d}t = \eta \int_0^T \dot{q}_t^2 \,\mathrm{d}t.$$

Variance. By Itô's isometry:

$$\mathrm{Var}(\Pi_{\mathrm{TC}}) = \sigma^2 \int_0^T (q_t - Q)^2 \,\mathrm{d}t.$$

4.4 Variational formulation

$$J_{\mathrm{TC}}[q] = \eta \int_0^T \dot{q}_t^2 \,\mathrm{d}t + \lambda\sigma^2 \int_0^T (q_t - Q)^2 \,\mathrm{d}t, \tag{7}$$

subject to $q_0 = Q$, $q_T = 0$.

Compared to IS, the risk term now penalises the deviation of q_t from Q rather than from 0. This makes sense: for a TC order, risk arises from having *sold too early*, because we are short relative to the benchmark S_T. The optimal strategy therefore *delays* selling compared to IS.

4.5 Euler–Lagrange equation

The Lagrangian is $\mathcal{L}_{\mathrm{TC}}(q, \dot{q}) = \eta \dot{q}^2 + \lambda \sigma^2 (q - Q)^2$. Computing the partial derivatives:

$$\frac{\partial \mathcal{L}_{\mathrm{TC}}}{\partial \dot{q}} = 2\eta \dot{q}, \qquad \frac{\partial \mathcal{L}_{\mathrm{TC}}}{\partial q} = 2\lambda \sigma^2 (q - Q).$$

The Euler–Lagrange equation gives:

$$\ddot{q} - \kappa^2 (q - Q) = 0, \qquad \kappa^2 = \frac{\lambda \sigma^2}{\eta}.$$

4.6 Change of variable

Define $y(t) = q(t) - Q$, which measures how far we are from a full position. Then $\ddot{y} = \ddot{q}$, and the ODE becomes:

$$\ddot{y} - \kappa^2 y = 0.$$

The boundary conditions translate to:

$$y(0) = q(0) - Q = Q - Q = 0, \qquad y(T) = q(T) - Q = 0 - Q = -Q.$$

4.7 Solution

General solution: $y(t) = A \cosh(\kappa t) + B \sinh(\kappa t)$.

At $t = 0$: $y(0) = A = 0$.

At $t = T$: $B \sinh(\kappa T) = -Q$, so $B = -Q / \sinh(\kappa T)$.

Therefore $y(t) = -Q \sinh(\kappa t) / \sinh(\kappa T)$, and recalling $q(t) = Q + y(t)$:

$$q_{\mathrm{TC}}(t) = Q \left(1 - \frac{\sinh(\kappa t)}{\sinh(\kappa T)} \right), \qquad v_{\mathrm{TC}}(t) = Q \, \frac{\kappa \cosh(\kappa t)}{\sinh(\kappa T)}, \qquad \kappa^2 = \frac{\lambda \sigma^2}{\eta}.$$

The schedule starts slowly (holding most of the position) and accelerates toward the end — the mirror image of IS.

5 TWAP Benchmark

5.1 Benchmark and payoff

The *time-weighted average price* benchmark is the time-average of the mid-price:

$$B_{\text{TWAP}} = \frac{1}{T}\int_0^T S_t \,\mathrm{d}t.$$

The payoff is:

$$\Pi_{\text{TWAP}} = X_T - Q \cdot B_{\text{TWAP}} = X_T - \frac{Q}{T}\int_0^T S_t \,\mathrm{d}t.$$

5.2 Handling the benchmark term

We need to simplify $\frac{Q}{T}\int_0^T S_t \,\mathrm{d}t$. Substituting $S_t = S_0 + \sigma W_t$:

$$\frac{Q}{T}\int_0^T S_t \,\mathrm{d}t = \frac{Q}{T}\int_0^T S_0 \,\mathrm{d}t + \frac{Q\sigma}{T}\int_0^T W_t \,\mathrm{d}t = QS_0 + \frac{Q\sigma}{T}\int_0^T W_t \,\mathrm{d}t.$$

We now simplify the remaining integral $\int_0^T W_t \,\mathrm{d}t$ using integration by parts. Set $u = W_t$ and $\mathrm{d}V = \mathrm{d}t$, so that $\mathrm{d}u = \mathrm{d}W_t$ and $V = t$:

$$\int_0^T W_t \,\mathrm{d}t = \Big[t\, W_t\Big]_0^T - \int_0^T t \;\mathrm{d}W_t = TW_T - \int_0^T t \;\mathrm{d}W_t. \tag{8}$$

Writing $W_T = \int_0^T \mathrm{d}W_t$, the first term becomes $TW_T = T\int_0^T \mathrm{d}W_t = \int_0^T T \;\mathrm{d}W_t$, so the two terms combine into a single stochastic integral:

$$\int_0^T W_t \,\mathrm{d}t = \int_0^T (T-t) \;\mathrm{d}W_t.$$

Therefore:

$$\frac{Q}{T}\int_0^T S_t \,\mathrm{d}t = QS_0 + \frac{Q\sigma}{T}\int_0^T (T-t) \;\mathrm{d}W_t.$$

Define the *deterministic linear schedule*

$$f(t) := Q\,\frac{T-t}{T},$$

which starts at Q when $t = 0$ and reaches 0 at $t = T$: it is the inventory path of a trader selling at a perfectly uniform rate Q/T. Equation (**??**) shows that the stochastic risk of a TWAP order is entirely governed by the *deviation* $q_t - f(t)$. This requires no time-average of q — the benchmark enters directly as a path.

5.3 Mean and variance

Mean. The stochastic integral has zero expectation:

$$-\mathbb{E}[\Pi_{\text{TWAP}}] = \eta\int_0^T v_t^2 \,\mathrm{d}t = \eta\int_0^T \dot{q}_t^2 \,\mathrm{d}t.$$

Variance. By Itô's isometry applied to (**??**):

$$\mathrm{Var}(\Pi_{\mathrm{TWAP}}) = \sigma^2 \int_0^T \left(q_t - Q\,\frac{T-t}{T} \right)^2 \mathrm{d}t.$$

5.4 Variational formulation

$$J_{\mathrm{TWAP}}[q] = \eta \int_0^T \dot{q}_t^2 \,\mathrm{d}t + \lambda\sigma^2 \int_0^T \left(q_t - Q\,\frac{T-t}{T} \right)^2 \mathrm{d}t, \tag{9}$$

subject to $q_0 = Q$, $q_T = 0$.

The three objectives have identical structure:

$$J[q] = \eta \int_0^T \dot{q}_t^2 \,\mathrm{d}t + \lambda\sigma^2 \int_0^T \big(q_t - f(t)\big)^2 \,\mathrm{d}t,$$

where the "target path" f changes by benchmark:

Benchmark	Target path $f(t)$	Intuition
IS	0	Hold nothing; sell everything up front.
TC	Q	Hold everything; sell everything at the end.
TWAP	$Q(T-t)/T$	Hold and sell at a constant linear rate.

5.5 Euler–Lagrange equation

The Lagrangian is $\mathcal{L}_{\mathrm{TWAP}}(q, \dot{q}, t) = \eta\dot{q}^2 + \lambda\sigma^2(q - f(t))^2$, where $f(t) = Q(T-t)/T$ is a *known, deterministic* function of time. Computing the partial derivatives:

$$\frac{\partial \mathcal{L}_{\mathrm{TWAP}}}{\partial \dot{q}} = 2\eta\dot{q}, \qquad \frac{\partial \mathcal{L}_{\mathrm{TWAP}}}{\partial q} = 2\lambda\sigma^2(q - f(t)).$$

The Euler–Lagrange equation gives:

$$\ddot{q} - \kappa^2 \left(q - Q\,\frac{T-t}{T} \right) = 0, \qquad \kappa^2 = \frac{\lambda\sigma^2}{\eta}.$$

5.6 Change of variable

Define the deviation from the linear schedule:

$$y(t) \coloneqq q(t) - f(t) = q(t) - Q\,\frac{T-t}{T}.$$

Since f is linear, its second derivative vanishes: $\ddot{f}(t) = 0$. Therefore $\ddot{y} = \ddot{q}$, and the ODE reduces to:

$$\ddot{y} - \kappa^2 y = 0.$$

Crucially, this is the same ODE as before — but the boundary conditions are now:

$$y(0) = q(0) - f(0) = Q - Q = 0, \qquad y(T) = q(T) - f(T) = 0 - 0 = 0.$$

Both boundary values are zero!

5.7 Solution

General solution: $y(t) = A\cosh(\kappa t) + B\sinh(\kappa t)$.

At $t = 0$: $y(0) = A = 0$.

At $t = T$: $B\sinh(\kappa T) = 0$.

Since $\kappa > 0$, we have $\sinh(\kappa T) > 0$, which forces $B = 0$.

Therefore $y(t) = 0$ for all $t \in [0, T]$, meaning $q(t) = f(t)$ exactly:

$$q_{\text{TWAP}}(t) = Q\left(1 - \frac{t}{T}\right), \qquad v_{\text{TWAP}}(t) = \frac{Q}{T}.$$

The optimal strategy is to sell at a **perfectly constant rate** Q/T for all t, for **any** value of $\lambda > 0$ and any κ. This is an exact result, not an approximation.

Why the result is exact. The function $f(t) = Q(T - t)/T$ already satisfies both boundary conditions $f(0) = Q$ and $f(T) = 0$, so it is a *feasible* trajectory. Since the objective (9) is a sum of squares of $\dot{q}_t$ and $(q_t - f(t))$, it is minimised at zero when $q_t = f(t)$: the deviation $y(t) = 0$ simultaneously zeroes out the risk term. There is no trade-off to resolve — selling linearly at the TWAP rate is both the cheapest (in impact) and the safest (in risk) thing to do within the TWAP benchmark.

6 Summary

All three strategies share the same Euler–Lagrange structure, $\ddot{y} - \kappa^2 y = 0$ on the deviation $y(t) = q(t) - f(t)$, with $\kappa^2 = \lambda\sigma^2/\eta$.

Benchmark	**Optimal inventory** $q(t)$	**BCs on** y	B
IS	$Q\dfrac{\sinh\kappa(T-t)}{\sinh\kappa T}$	$y(0) = Q,\ y(T) = 0$	$-Q\coth\kappa T$
TC	$Q\left(1 - \dfrac{\sinh\kappa t}{\sinh\kappa T}\right)$	$y(0) = 0,\ y(T) = -Q$	$-\dfrac{Q}{\sinh\kappa T}$
TWAP	$Q\left(1 - \dfrac{t}{T}\right)$	$y(0) = 0,\ y(T) = 0$	0

The stochastic risk of each strategy, identified from the payoff via integration by

parts and Itô's isometry, is:

$$\mathrm{Var}(\Pi) = \sigma^2 \int_0^T \big(q_t - f(t)\big)^2 \,\mathrm{d}t, \quad \text{where } f(t) = \begin{cases} 0 & \text{(IS)} \\ Q & \text{(TC)} \\ Q(T-t)/T & \text{(TWAP)} \end{cases}$$

7 Efficient Frontiers for IS and TC

7.1 Implementation Shortfall

Cost. Recall the optimal IS trading speed:

$$v_{\mathrm{IS}}(t) = Q\,\frac{\kappa\cosh\big(\kappa(T-t)\big)}{\sinh(\kappa T)}.$$

We compute:

$$\mathrm{Cost}_{\mathrm{IS}} = \eta \int_0^T v_{\mathrm{IS}}(t)^2 \,\mathrm{d}t = \eta Q^2 \frac{\kappa^2}{\sinh^2(\kappa T)} \int_0^T \cosh^2\big(\kappa(T-t)\big)\,\mathrm{d}t.$$

Substituting $u = T - t$:

$$\int_0^T \cosh^2\big(\kappa(T-t)\big)\,\mathrm{d}t = \int_0^T \cosh^2(\kappa u)\,\mathrm{d}u.$$

Using the identity $\cosh^2(\kappa u) = \dfrac{1+\cosh(2\kappa u)}{2}$:

$$\int_0^T \cosh^2(\kappa u)\,\mathrm{d}u = \frac{T}{2} + \frac{1}{2}\int_0^T \cosh(2\kappa u)\,\mathrm{d}u = \frac{T}{2} + \frac{\sinh(2\kappa T)}{4\kappa}.$$

Using the identity $\sinh(2\kappa T) = 2\sinh(\kappa T)\cosh(\kappa T)$:

$$\int_0^T \cosh^2(\kappa u)\,\mathrm{d}u = \frac{T}{2} + \frac{\sinh(\kappa T)\cosh(\kappa T)}{2\kappa}.$$

Therefore:

$$\begin{aligned}
\mathrm{Cost}_{\mathrm{IS}} &= \frac{\eta Q^2 \kappa^2}{\sinh^2(\kappa T)} \left(\frac{T}{2} + \frac{\sinh(\kappa T)\cosh(\kappa T)}{2\kappa} \right) \\
&= \frac{\eta Q^2 \kappa^2}{2\sinh^2(\kappa T)} \left(T + \frac{\sinh(\kappa T)\cosh(\kappa T)}{\kappa} \right) \\
&= \frac{\eta Q^2 \kappa}{2} \left(\frac{\kappa T}{\sinh^2(\kappa T)} + \frac{\cosh(\kappa T)}{\sinh(\kappa T)} \right) \\
&= \frac{\eta Q^2 \kappa}{2} \left(\frac{\kappa T}{\sinh^2(\kappa T)} + \coth(\kappa T) \right).
\end{aligned}$$

$$\text{Cost}_{\text{IS}} = \frac{\eta Q^2 \kappa}{2} \left(\coth(\kappa T) + \frac{\kappa T}{\sinh^2(\kappa T)} \right).$$

Risk. Recall the optimal IS inventory:

$$q_{\text{IS}}(t) = Q \, \frac{\sinh\big(\kappa(T-t)\big)}{\sinh(\kappa T)}.$$

We compute:

$$\text{Risk}_{\text{IS}} = \sigma^2 \int_0^T q_{\text{IS}}(t)^2 \, \mathrm{d}t = \sigma^2 Q^2 \, \frac{1}{\sinh^2(\kappa T)} \int_0^T \sinh^2\big(\kappa(T-t)\big) \, \mathrm{d}t.$$

Substituting $u = T - t$:

$$\int_0^T \sinh^2\big(\kappa(T-t)\big) \, \mathrm{d}t = \int_0^T \sinh^2(\kappa u) \, \mathrm{d}u.$$

Using the identity $\sinh^2(\kappa u) = \dfrac{\cosh(2\kappa u) - 1}{2}$:

$$\int_0^T \sinh^2(\kappa u) \, \mathrm{d}u = \frac{1}{2} \int_0^T \cosh(2\kappa u) \, \mathrm{d}u - \frac{T}{2} = \frac{\sinh(2\kappa T)}{4\kappa} - \frac{T}{2}.$$

Using $\sinh(2\kappa T) = 2\sinh(\kappa T)\cosh(\kappa T)$:

$$\int_0^T \sinh^2(\kappa u) \, \mathrm{d}u = \frac{\sinh(\kappa T)\cosh(\kappa T)}{2\kappa} - \frac{T}{2}.$$

Therefore:

$$\begin{aligned}
\text{Risk}_{\text{IS}} &= \frac{\sigma^2 Q^2}{\sinh^2(\kappa T)} \left(\frac{\sinh(\kappa T)\cosh(\kappa T)}{2\kappa} - \frac{T}{2} \right) \\
&= \frac{\sigma^2 Q^2}{2\kappa} \left(\frac{\cosh(\kappa T)}{\sinh(\kappa T)} - \frac{\kappa T}{\sinh^2(\kappa T)} \right) \\
&= \frac{\sigma^2 Q^2}{2\kappa} \left(\coth(\kappa T) - \frac{\kappa T}{\sinh^2(\kappa T)} \right).
\end{aligned}$$

$$\text{Risk}_{\text{IS}} = \frac{\sigma^2 Q^2}{2\kappa} \left(\coth(\kappa T) - \frac{\kappa T}{\sinh^2(\kappa T)} \right).$$

7.2 Target Close

Cost. Recall the optimal TC trading speed:

$$v_{\text{TC}}(t) = Q \, \frac{\kappa \cosh(\kappa t)}{\sinh(\kappa T)}.$$

We compute:

$$\mathrm{Cost}_{\mathrm{TC}} = \eta \int_0^T v_{\mathrm{TC}}(t)^2 \, \mathrm{d}t = \eta Q^2 \frac{\kappa^2}{\sinh^2(\kappa T)} \int_0^T \cosh^2(\kappa t) \, \mathrm{d}t.$$

The integral $\int_0^T \cosh^2(\kappa t) \, \mathrm{d}t$ is identical to the one computed in the IS case (with $u = t$ instead of $u = T - t$):

$$\int_0^T \cosh^2(\kappa t) \, \mathrm{d}t = \frac{T}{2} + \frac{\sinh(\kappa T)\cosh(\kappa T)}{2\kappa}.$$

Therefore, the computation is word-for-word identical to the IS cost, giving:

$$\mathrm{Cost}_{\mathrm{TC}} = \frac{\eta Q^2 \kappa}{2} \left(\coth(\kappa T) + \frac{\kappa T}{\sinh^2(\kappa T)} \right) = \mathrm{Cost}_{\mathrm{IS}}.$$

Risk. For TC the risk integrand is $(q_t - Q)^2$. Recall:

$$q_{\mathrm{TC}}(t) - Q = -Q \frac{\sinh(\kappa t)}{\sinh(\kappa T)}.$$

We compute:

$$\mathrm{Risk}_{\mathrm{TC}} = \sigma^2 \int_0^T \left(q_{\mathrm{TC}}(t) - Q \right)^2 \, \mathrm{d}t = \sigma^2 Q^2 \frac{1}{\sinh^2(\kappa T)} \int_0^T \sinh^2(\kappa t) \, \mathrm{d}t.$$

Again, the integral $\int_0^T \sinh^2(\kappa t) \, \mathrm{d}t$ is identical to the one computed in the IS case:

$$\int_0^T \sinh^2(\kappa t) \, \mathrm{d}t = \frac{\sinh(\kappa T)\cosh(\kappa T)}{2\kappa} - \frac{T}{2}.$$

Therefore:

$$\mathrm{Risk}_{\mathrm{TC}} = \frac{\sigma^2 Q^2}{2\kappa} \left(\coth(\kappa T) - \frac{\kappa T}{\sinh^2(\kappa T)} \right) = \mathrm{Risk}_{\mathrm{IS}}.$$

7.3 Why the Frontiers Coincide

The equality of cost and risk for IS and TC is not a coincidence. The two schedules are related by a time-reversal:

$$q_{\mathrm{TC}}(t) = Q - q_{\mathrm{IS}}(T - t).$$

Taking the derivative: $v_{\mathrm{TC}}(t) = v_{\mathrm{IS}}(T - t)$, so the cost integrands are related by $t \mapsto T - t$ and integrate to the same value. For the risk, note that

$$q_{\mathrm{TC}}(t) - Q = -q_{\mathrm{IS}}(T - t),$$

so the risk integrands are also related by $t \mapsto T - t$ and again integrate identically.

The IS and TC efficient frontiers coincide exactly:

$$\mathrm{Cost}^\star(\kappa) = \frac{\eta Q^2 \kappa}{2}\left(\coth(\kappa T) + \frac{\kappa T}{\sinh^2(\kappa T)}\right)$$

$$\mathrm{Risk}^\star(\kappa) = \frac{\sigma^2 Q^2}{2\kappa}\left(\coth(\kappa T) - \frac{\kappa T}{\sinh^2(\kappa T)}\right),$$

for all $\kappa = \sqrt{\lambda\sigma^2/\eta} > 0$.